

Interlocking Sustainable Foam-Core Chair

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Abstract—A sustainable foam-core chair was designed, analyzed, and tested to meet dimensional and weight requirements while supporting a 103 kg load without fasteners. The goal was to demonstrate lightweight structural design using foam-core to enable low-waste prototyping during the development process. Foam-core, though not recyclable, enables quick and resource-efficient prototyping that exposes structural and ergonomic issues early, guiding more sustainable manufacturing. Reducing waste is essential as it lowers material consumption, decreases environmental impact, and improves the overall efficiency of the design cycle. Three concept designs were made and evaluated using a Pugh chart, and the winning concept was modeled in Inventor. Analysis was done using lines of force sketches along with FEA to find and remove stress concentration. The final CAD model properties predicted 462 g as the weight of the physical chair. A physical prototype was made using a laser cutter and assembled using interlocking parts. The chair was tested incrementally, using lighter to heavier people sitting on the assembled chair, ending with the 103 kg person, where minimal damage occurred. The results demonstrated that careful load-bearing design, light weighting, and targeted reinforcement can produce a sustainable foam-core chair suitable for real-world use.

I. INTRODUCTION

A. Motivation and Context

Sustainable product design reduces material waste and simplifies end-of-life recycling, providing both environmental and economic benefits. Furniture is a common consumer product, and sustainable design approaches can be beneficial. Normally, furniture, especially chairs, is made of wood or plastic which is joined by fasteners, all of which increase price, waste, and make it hard to dispose of. This report explores a design of a lightweight chair made of foam core board, without the use of adhesives or fasteners. Eliminating adhesives and fasteners reduces waste and simplifies end-of-life recycling, making it a relevant case study in sustainable product development. In order to reduce products detrimental environmental impacts, it is important to consider new ways of making products more sustainable.

The design (Fig. 1) demonstrates light weighting principles commonly used in mechanical and aerospace industries, demonstrating how the engineering process can be beneficial for product design. This concept chair is a lightweight, recyclable chair that maintains strength and function. To explore these concepts in practice, this project applies sustainable design principles to develop a sustainable foam-core chair.

B. Project Objectives and Functional Requirements

The primary objective of the project was to design and fabricate a foam-core chair capable of supporting a 103 kg load

while meeting dimensional and sustainability criteria. Other functional requirements include:

- Seat height: 480 mm to 530 mm
- Seat depth: 410 mm to 450 mm
- Seat width: 410 mm to 450 mm
- Back height (from seat): 280 mm to 480 mm
- Material: foam core board sheets (457 mm × 610 mm × 5 mm).
- No fasteners, adhesives, or tape.
- Target mass < 800 g, with preference toward < 400 g.



Fig. 1: Final Prototype

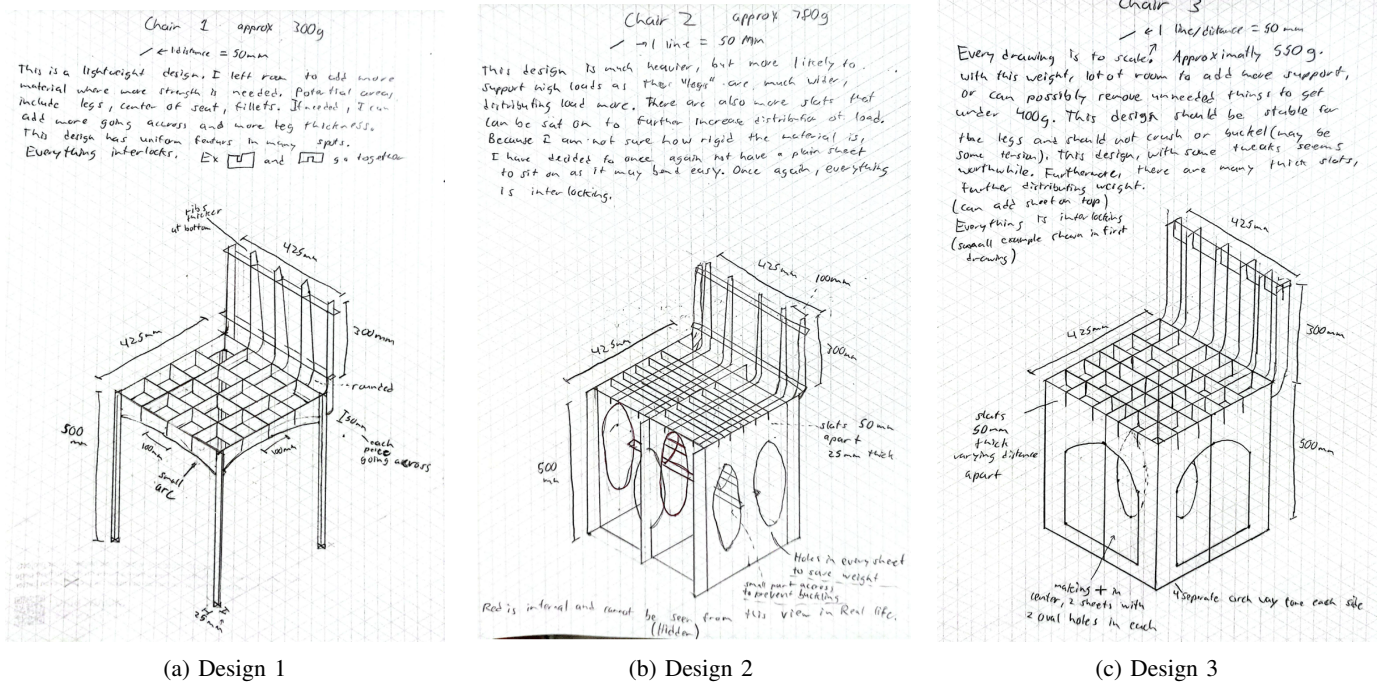


Fig. 2: Hand-Sketched Concepts

II. CONCEPT GENERATION AND EVALUATION

A. Sketches and Concept Descriptions

With the project goals defined, the next phase focused on generating and evaluating multiple chair concepts to identify the most effective design. Three design sketches were created to give different ideas to the chair, Designs 1-3 (Fig. 2). All of the designs featured backrests that starts with the slats, then comes up to act as the back rest, where then all of the ribs for the back rest are connected for stability. Each design included slats, however, some had more or less than others. Similarly, the designs all included interlocking slots and members to hold together.

Design 1 was a extra lightweight concept targeted to be under 400 g. It included small legs in each corner and chair slats to help distribute load evenly. The legs were skinny, and small arched pieces attached each leg together to help avoid high stress areas.

Design 2 had 5 "legs", which were large pieces for foam core which had holes cutout to reduce weight. This design had more slats than the first one, but shorter ones to help distribute weight more while keeping the weight down.

Design 3 was similar to Design 2, but it had slightly different cutouts, leg layout, and more slats.

Design 1, being the lightest, had the highest stress concentration, while Design 3 distributed stress more evenly. To objectively compare the three designs, a Pugh chart was used for concept evaluation.

B. Pugh Chart and Decision Rationale

TABLE I: Pugh chart used to evaluate concepts.

Criteria	Weight	Design 1	Design 2	Design 3
Dimensions	4	0	0	0
Chair Weight	4	0	-	-
Looks	1	0	0	0
Comfort	1	0	0	+
Ease of Build	2	0	-	0
Ease of Design	2	0	-	0
Number of Parts	3	0	0	0
Fasteners	4	0	0	0
Chance of Success	6	0	+	+
Weighted Total		0	-2	3

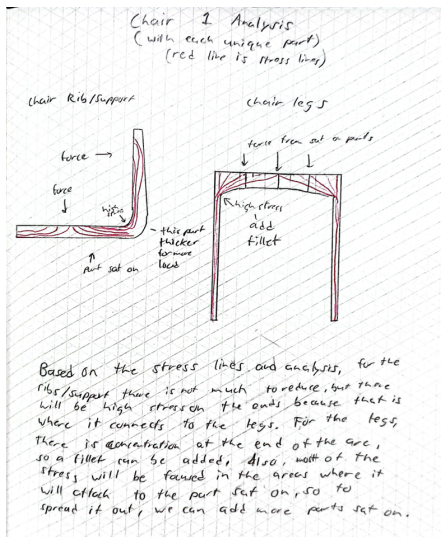
The Pugh chart (Table 1) provided a structured framework for comparing concepts based on weighted design criteria, using Design 1 as the baseline. The Pugh chart analysis provided quantitative insight into the relative strengths of each concept, guiding the final design choice. The analysis indicated that Design 3 would be the most promising, as it achieved the highest weighted score.

Design 3 was selected because it had a high chance of success while also being the most comfortable, and not being too difficult to design or put together. Though it was not the lightest, due to its probability of success, it was selected over Design 1.

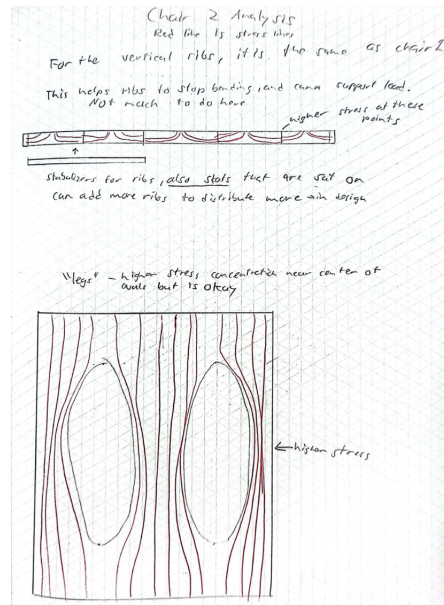
III. ANALYSIS

A. Lines of Force

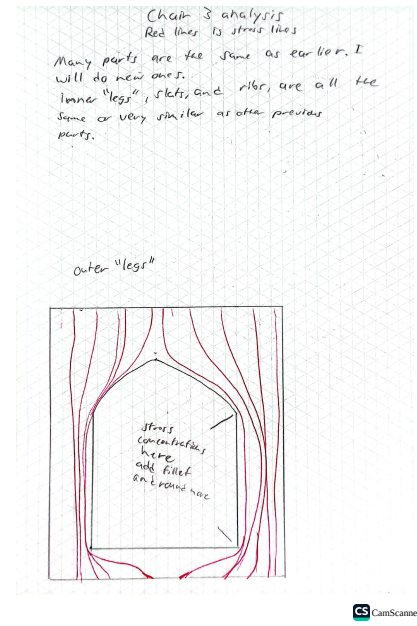
Following concept selection, analytical methods were used to predict stress concentrations and validate the structural



(a) Design 1

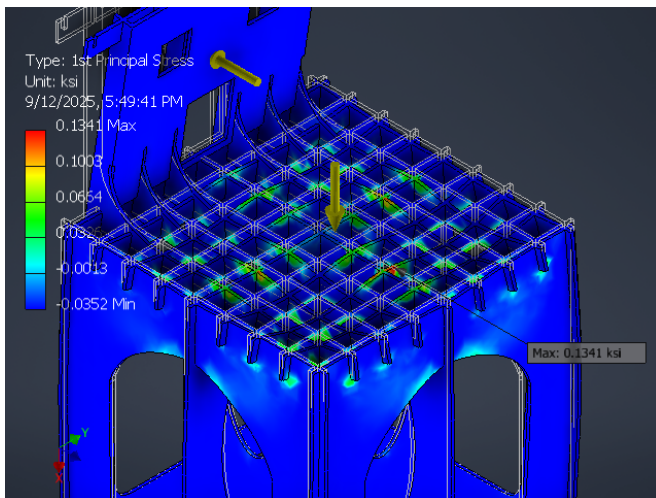


(b) Design 2

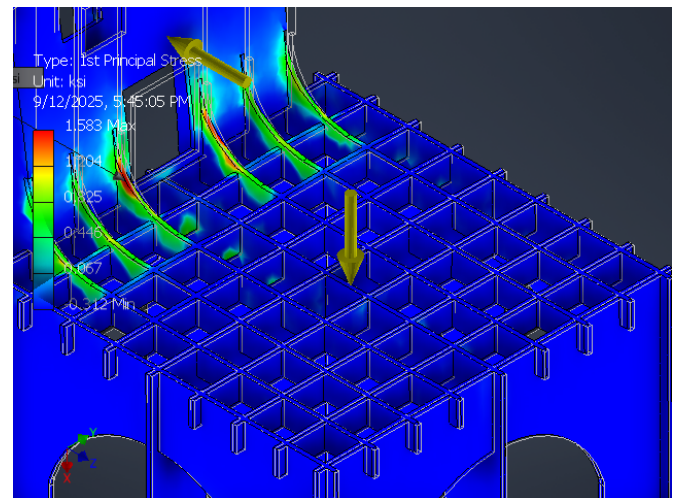


(c) Design 3

Fig. 3: Lines of Force



(a) Stress Analysis of All Weight On Seat



(b) Stress Analysis of Half Weight On Back

Fig. 4: FEA of the Chair Under Different Loads

integrity of the chosen design. The first method was lines of force (Fig. 3). For Design 3, high stress was found at the intersection of the slats, along with the arc of the ribs. Stress can be mitigated by adding more slats to distribute loads and increasing rib thickness near the corners. Design 3 had high stress concentration on the sides of the outer legs, along with on the sides and center of the inner legs. However, since stress levels remained within acceptable limits, no major geometry changes were required.

B. Finite Element Analysis

While the line-of-force sketches highlighted general load paths, finite element analysis provided a more detailed view of stress magnitudes and distributions. FEA was performed on the finalized Inventor model of Design 3 (Fig. 4). The analysis was done on the seat of the chair and the backrest. Figure 4 presents the von Mises stress results; the color scale follows the convention blue = low stress, green/yellow = moderate stress, and red = high stress (maximum). The results indicated high stress where each slat overlapped, along with where the slat connected to the legs. Significant stress concentrations were observed at the apex of the arches and at the front

corner of the backrest. These areas of stress concentrations were expected, and minor adjustments were made in order to reduce it. FEA of Design 3 confirmed stress concentration at slat intersections, leg joints, and rib arches, matching trends from line-of-force analysis.

C. Comparison: Lines of Force vs FEA

The hand sketches correctly predicted the global load paths (seat, ribs, and legs) and identified the same regions of stress flagged by FEA. Differences were in magnitude and localization of stresses: FEA showed peak stresses concentrated at small overlap regions and sharp corners that the hand sketches could not resolve. This comparison justified the use of small local geometry tweaks rather than larger global changes.

IV. EXPERIMENTAL RESULTS

A. Prototype Fabrication and Mass

After completing the analysis phase, a full-scale prototype was fabricated and tested to verify theoretical predictions. The prototype was fabricated by laser cutting foam core board sheets, based on DXF files created. The laser cutter made it easy to fabricate and assemble the chair and its interlocking parts as it cuts much more precise than by hand. Precision is needed for the parts to fit snug, so the chair is rigid, but not too snug where it is impossible to assemble. Originally, the assembled chair's mass was estimated to be 788 g, however, after getting the new density, it was estimated at 462 g, according to Inventor predictions. Initial laser cutting produced a subset of components, which were tested to verify fit and function prior to full-scale fabrication. From those simple tests and looking at stress analysis results, I made some tweaks to the design. After that, I was able to laser cut my parts and assemble the chair. During assembly, several components fractured due to tight tolerances; these were not modified due to time constraints.

B. Load Testing and Analysis

Once fabrication was complete, load testing was performed to evaluate the chair's actual strength and stability under increasing weight. Initial testing with a 70 kg load revealed that the relatively short and thin slats cracked under stress, though the main structure remained intact. Similar results were observed with 80 kg and 90 kg loads. Finally, a 103 kg load was used to test it, and it worked with little deformation to the main structure. The only deformation was slight bend in one of the four outside legs along with more of the slats breaking. Despite partial cracking of slats, the main frame supported a load exceeding 103 kg, validating the design requirements.

V. DISCUSSION

A. Meeting Requirements

The experimental results were then analyzed to assess whether the final design met the established functional and performance requirements. The design met all of the functional requirements and all of the dimensional constraints, which can be seen in the drawing of the assembly (see appendix).

The seat height was 490 mm, the seat depth and width were 450 mm and 445 mm, and the back height was 285 mm, all of which meet the requirements. The chair supported a 103 kg load while maintaining a total mass of 462 g, well below the 800 g target.

B. Light Weighting

To achieve the 462 g target, many light weighting techniques were used. For the back rest, ribs were used to provide the main structural support instead of solid boards, which would be less effective and heavier. The legs included arches used along with oval cutouts. The seat had slats instead of solid board, which increased strength and decreased weight. These features effectively reduced mass while maintaining stiffness.

C. Future Improvements

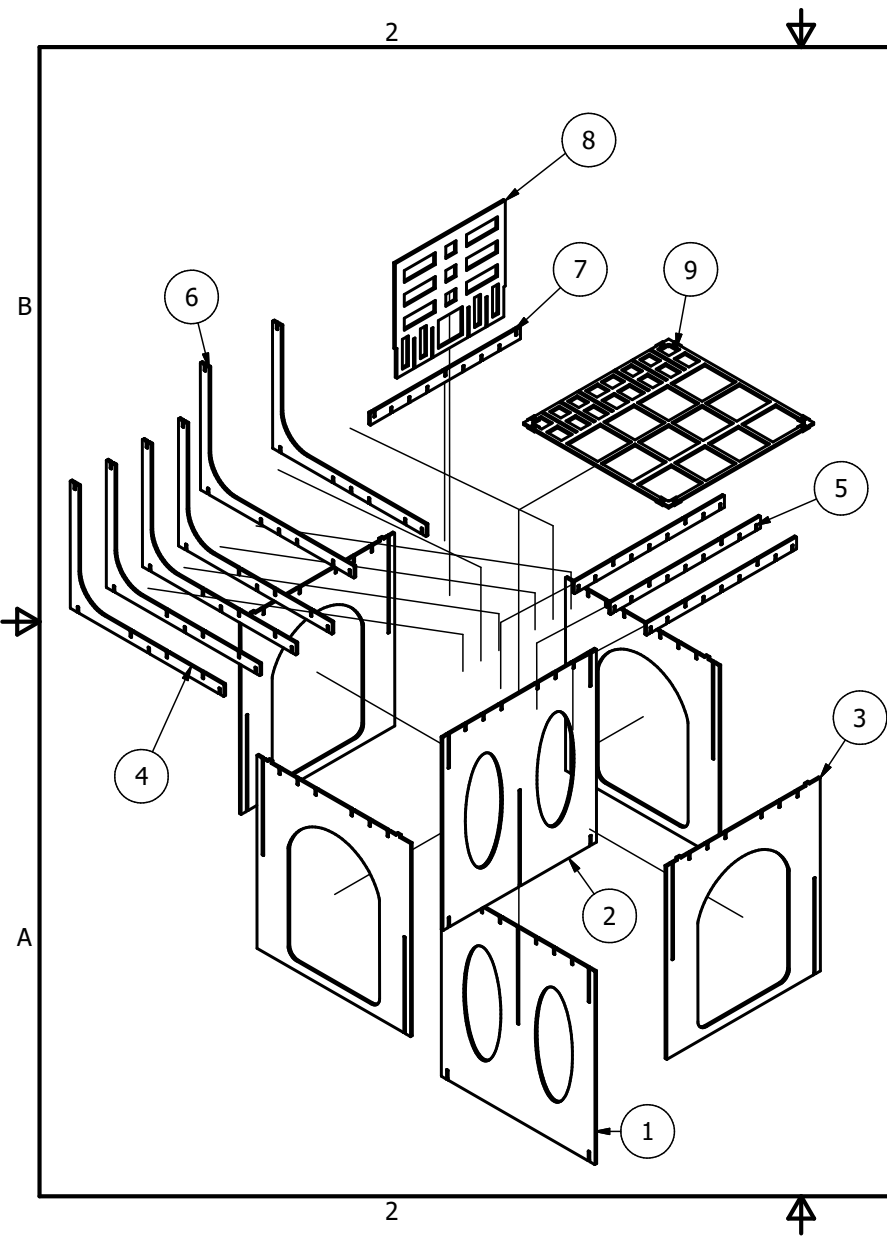
Although the lightweight design was effective, several areas for refinement were identified. Repositioning the main legs inward would improve load distribution. Also, weak point were created where the slats were interlocking as that part was thinner. Future iterations could employ non-interlocking slats arranged in alternating layers. Backrest reinforcement could be achieved by increasing the number and thickness of ribs in future iterations. These insights provide valuable guidance for future iterations and sustainable furniture development.

VI. CONCLUSION

This project was a success, as all of the objectives were achieved. A foam core board chair was designed, analyzed, fabricated, and tested to meet all of the requirements (dimensional, load support, weight limit and more). Three concept sketches were made and analyzed using lines of force sketches to find areas of high stress. A Pugh chart with certain weighted criteria was used to select the best concept. The chosen concept, Design 3, was modeled on Inventor and a FEA was done to further find stress concentration, in order to tweak the design accordingly. Inventor predicted a mass of 462 g, which is well in the acceptable range. The chair model, meeting all of the specifications, was then made into DXF files and laser cut, where some issues were found.

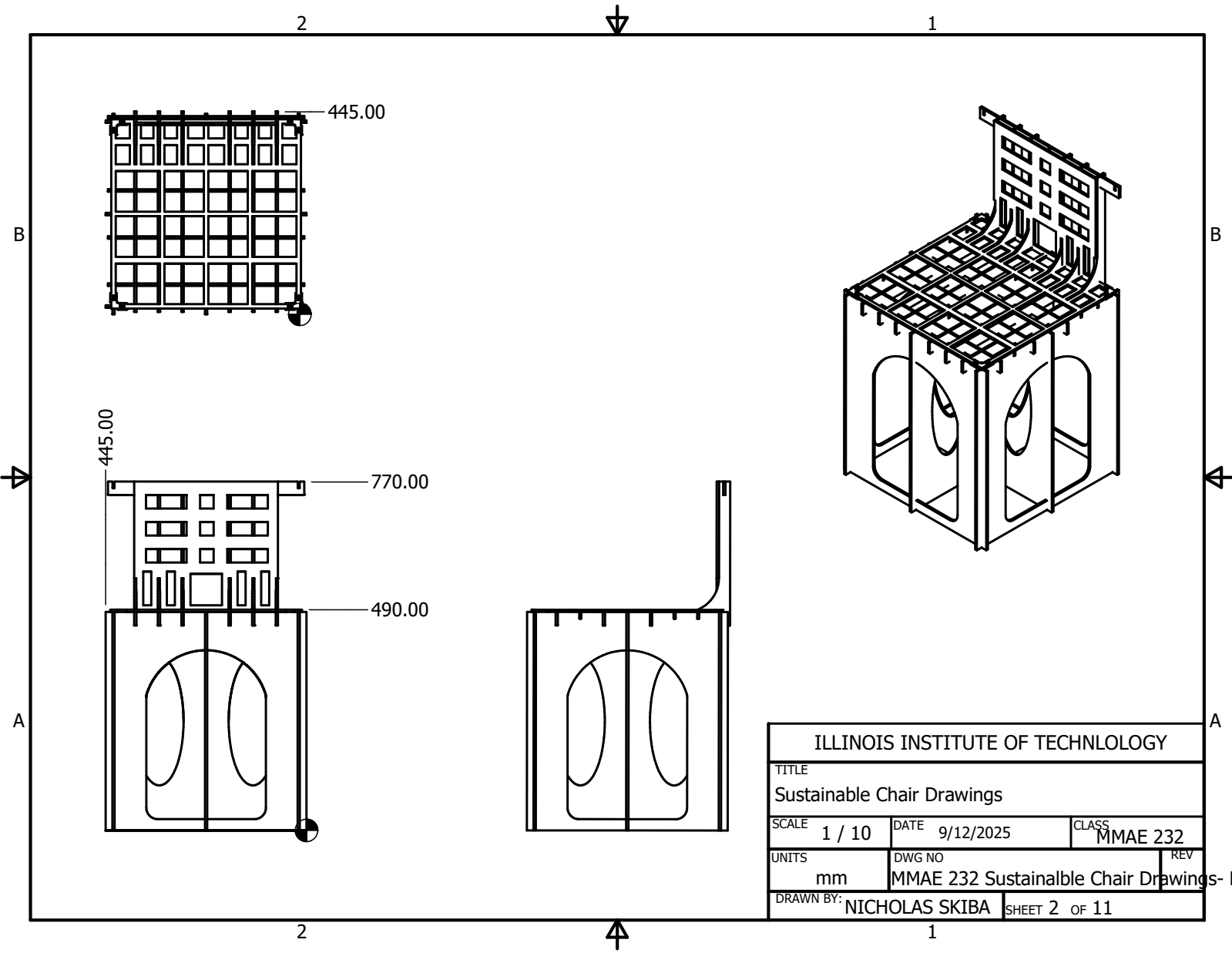
After minimal tweaks to the design, everything was laser cut and assembled. The interlocking structure and small tolerances led to a tight fit and some assembly issues, but the chair was overall successful, demonstrating high strength to weight ratio due to the chair being considerably lighter (223 times lighter) than the load it can hold. Future work could refine assembly tolerances and further reduce weight. Overall, the project demonstrated that sustainable, fastener-free furniture can be both lightweight and structurally sound.

APPENDIX A ENGINEERING DRAWINGS

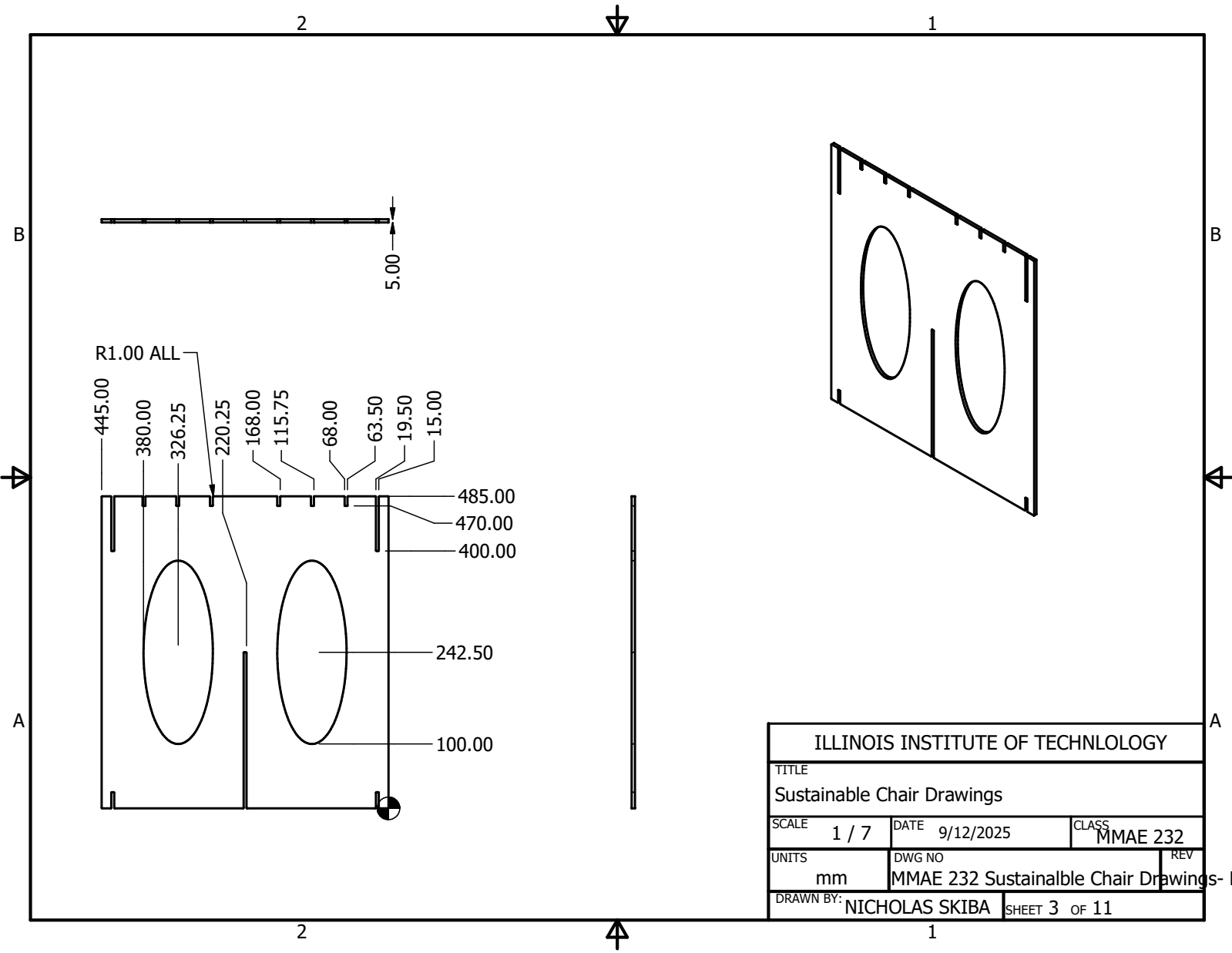


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3	4	MMAE 232 Sustainable Chair Outside Leg - Nicholas Skiba
4	4	MMAE 232 Sustainable Chair Slats- Nicholas Skiba
5	3	MMAE 232 Sustainable Chair Slats - Nicholas Skiba
6	2	MMAE 232 Sustainable Chair Slats And Back Over - Nicholas Skiba
7	1	MMAE 232 Sustainable Chair Slats Under - Nicholas Skiba
8	1	MMAE 232 Sustainable Chair Back Rest - Nicholas Skiba
9	1	MMAE 232 Sustainable Chair Seat - Nicholas Skiba

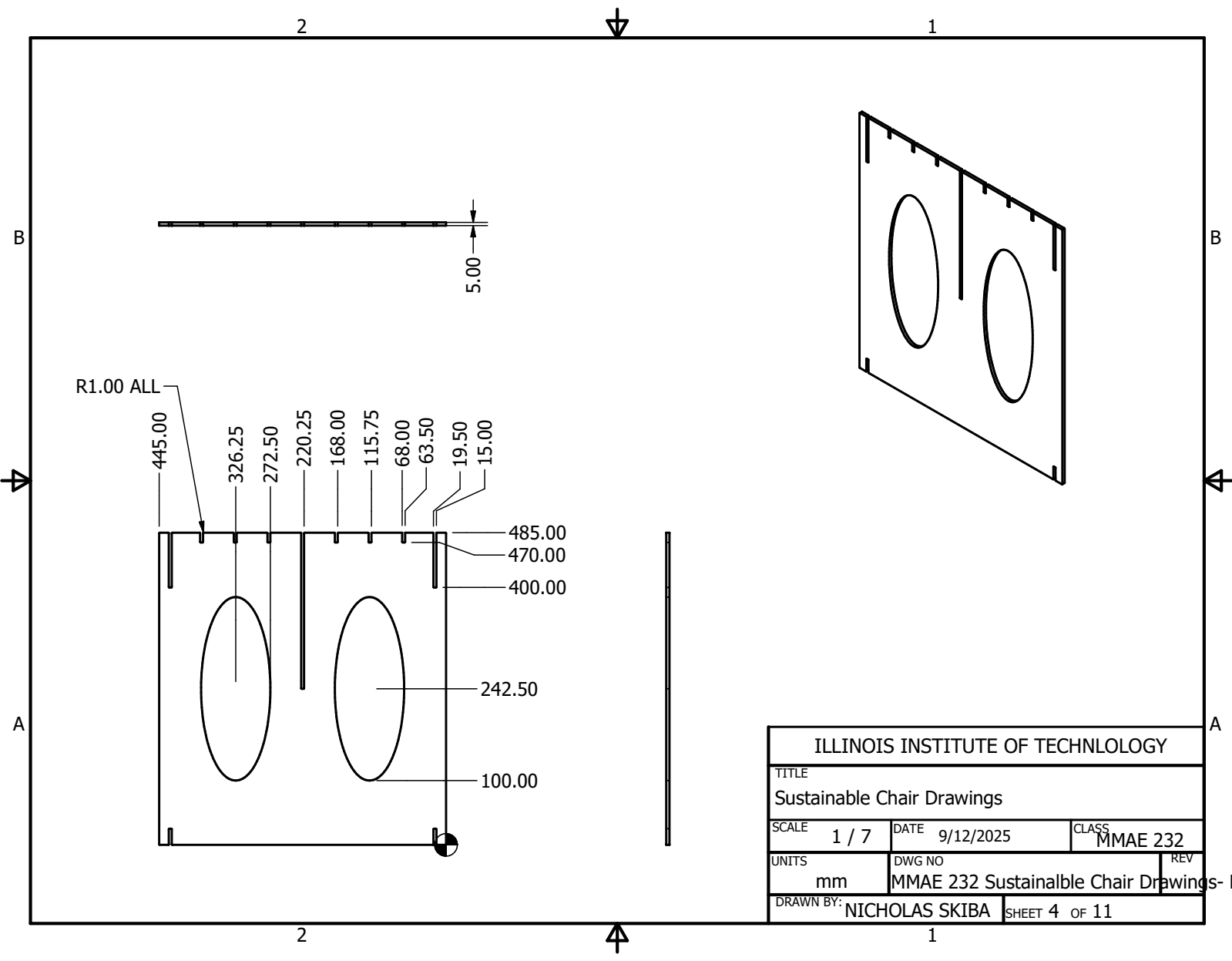
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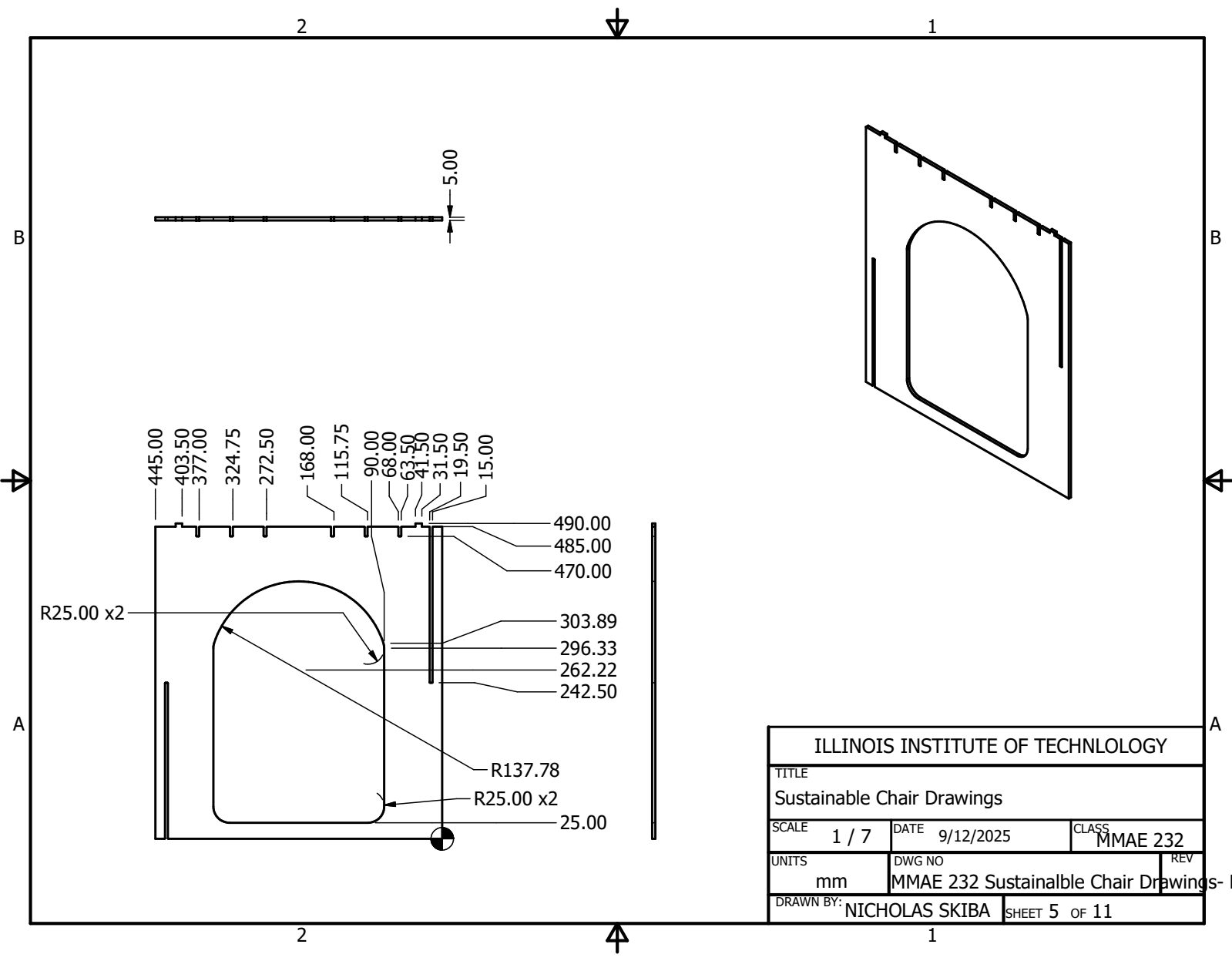


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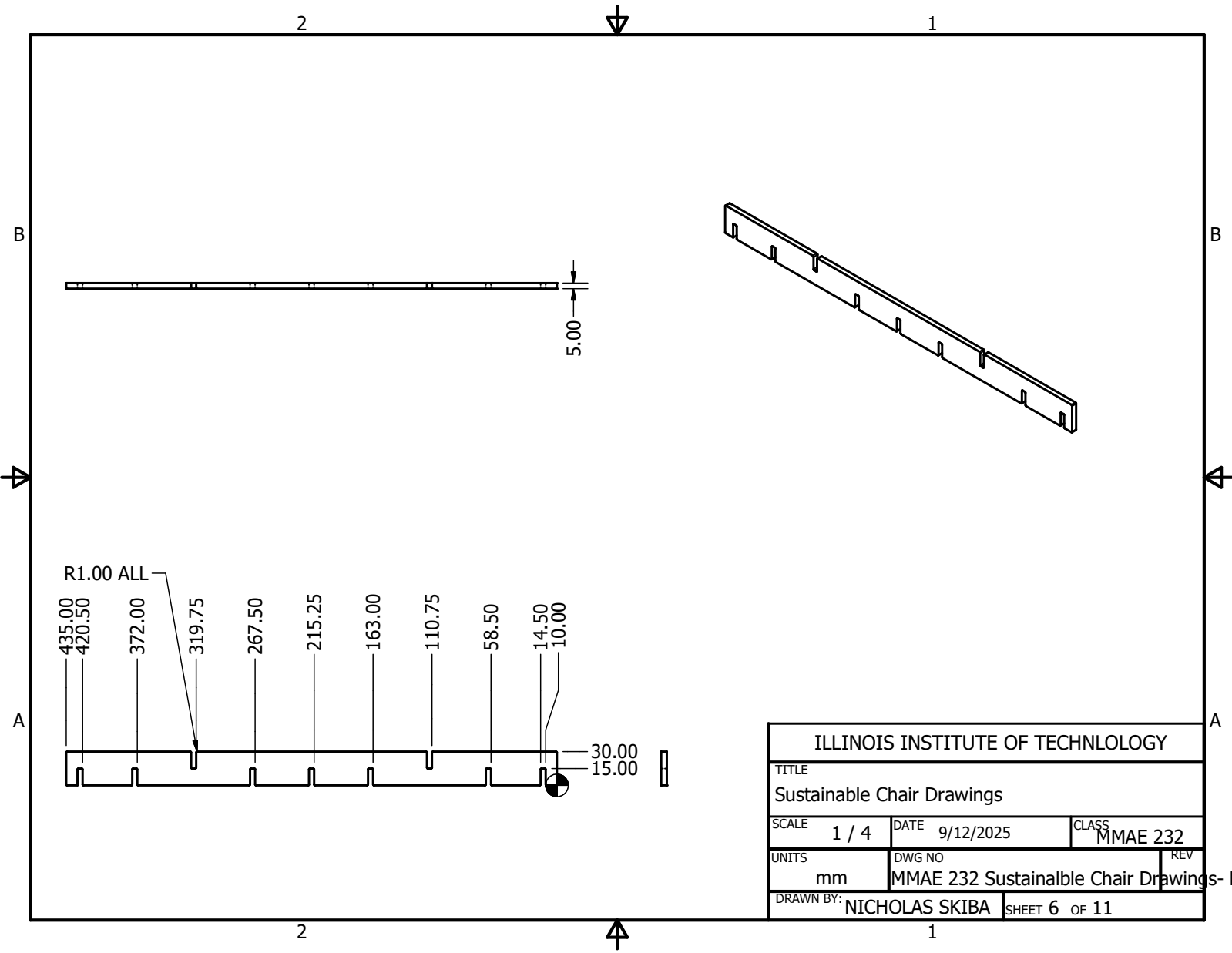


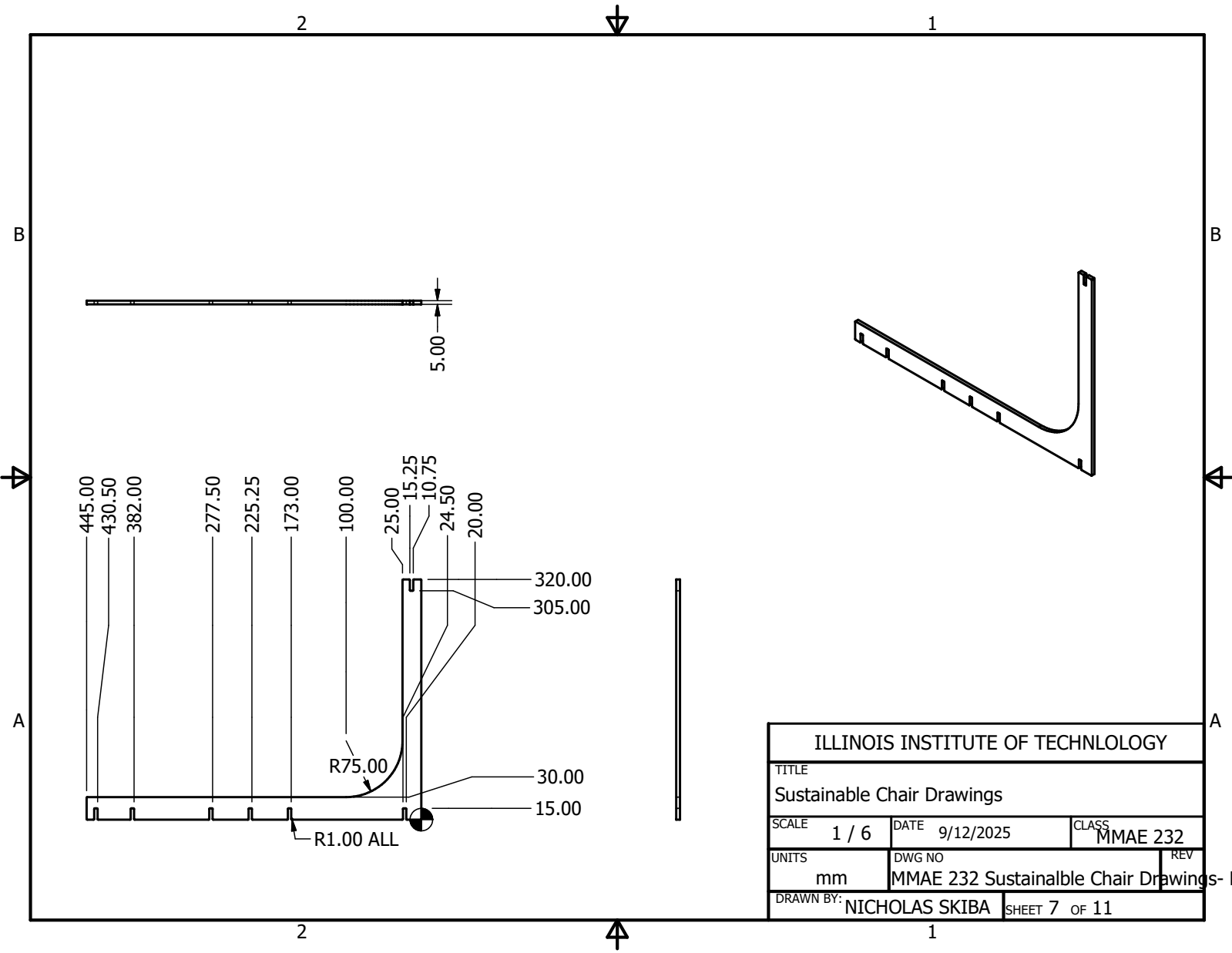
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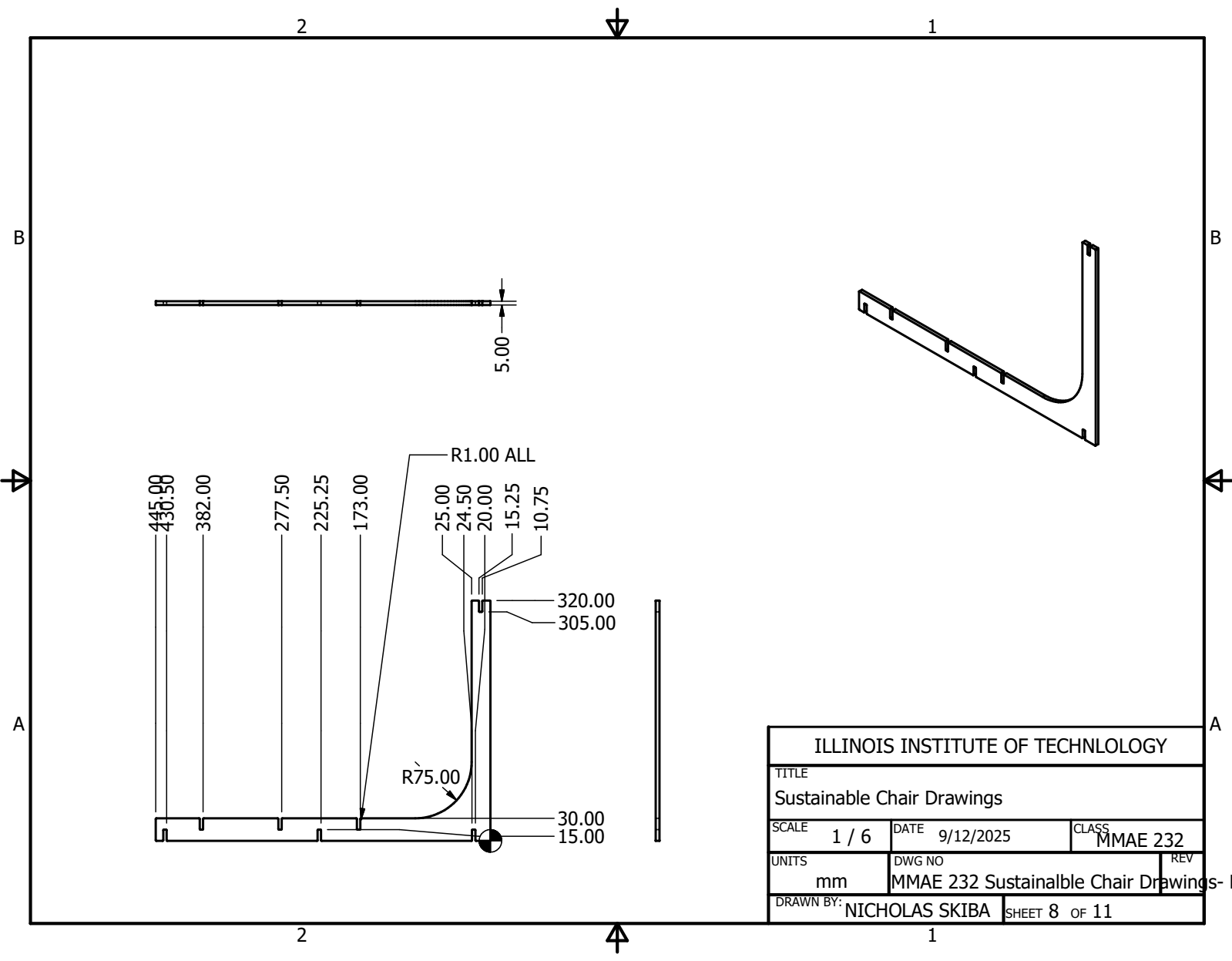


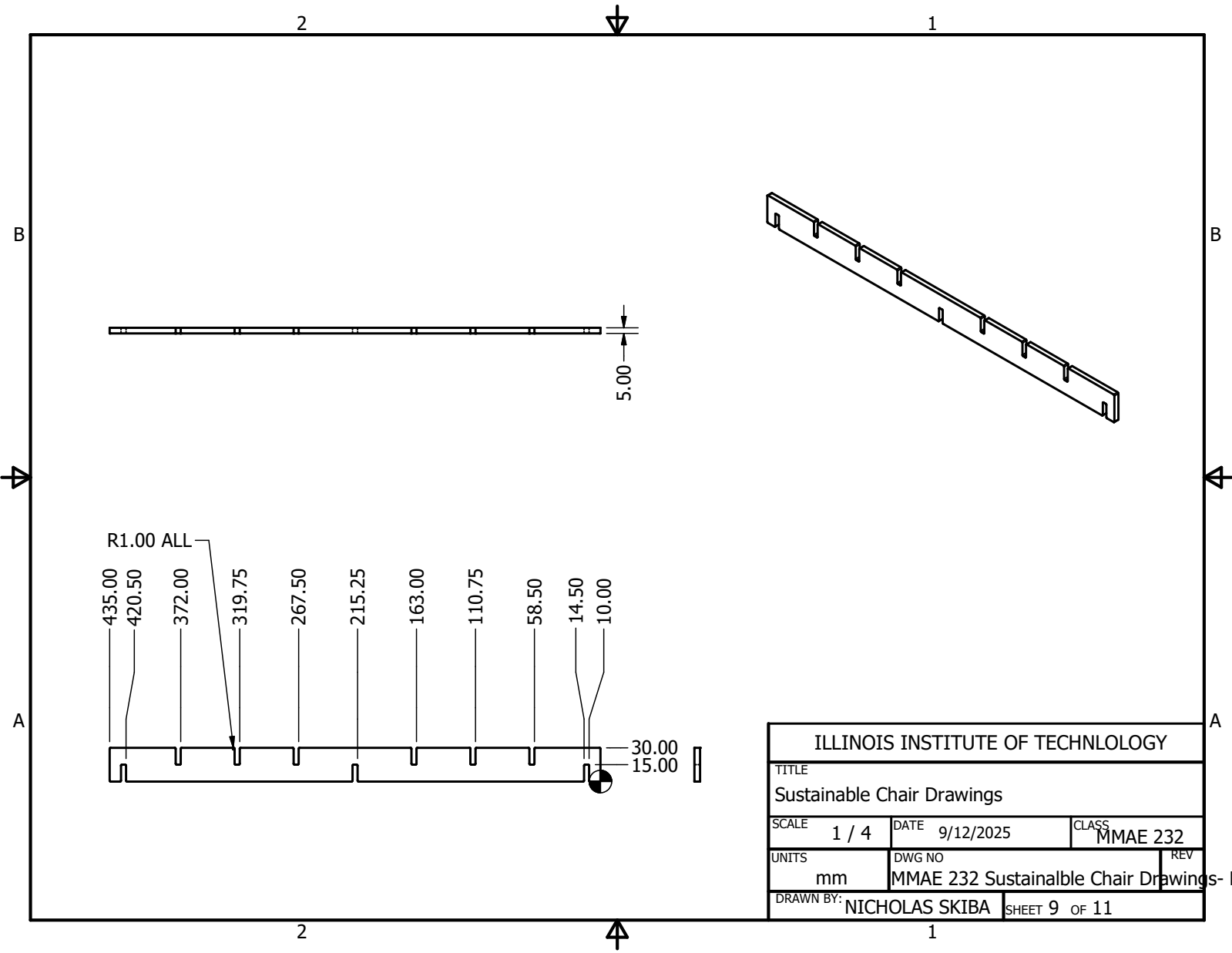


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